

Spring 2025. Physics 510.

Quantum Theory II.

Test # 2.

Show all your work step by step and write down your final answers explicitly. Plain final answers will not be graded. You are allowed to use your equation sheet. A4, one side, handwritten by you personally.

1. Consider a three-dimensional repulsive ($V_0 > 0$) square barrier of radius a .

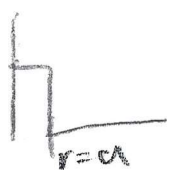
$$V(r) = \begin{cases} V_0, & r < a \\ 0, & r \geq a \end{cases}$$

A particle of energy $E = \frac{\hbar^2 k^2}{2m} < V_0$ is incident on the well.
Subj (csk)

- Write down the Schrödinger equation for an s-wave ($l = 0$ mode).
- Solve it directly for both regions (it is possible because $l = 0$). Do not forget that the asymptotically free solution must contain a partial phase shift.
- Derive the phase shift, δ_0 , for the mode by applying necessary boundary conditions.
- How does the phase shift behave if $V_0 \rightarrow \infty$?
- Derive the total cross-section of the mode, σ_0 , in the limit of zero energy.

2. Calculate the scattering amplitude and differential cross-section in the Born approximation for the same potential

$$V(r) = \begin{cases} V_0, & r < a \\ 0, & r \geq a \end{cases}$$

1)  $r=a$

$$\psi = \frac{e^{kr} - e^{-kr}}{2} + \frac{e^{kr} + e^{-kr}}{2} = e^{kr}$$

a) $r > a$ $-\frac{\hbar^2}{2m} \nabla^2 \psi = E \psi$ $k = \frac{\sqrt{2mE}}{\hbar}$

$r < a$ $-\frac{\hbar^2}{2m} \nabla^2 \psi + V_0 \psi = E \psi$ $\kappa^2 = \frac{2m(E - V_0)}{\hbar^2}$

b) $r > a$ $\psi = R_{k0} Y_{00} = R_{k0}$

$$\psi_{kr} = \alpha j_0(kr) + \beta n_0(kr) = C_0 (\cos \delta e^{i\delta} j_0(kr) - \sin \delta e^{i\delta} n_0(kr))$$

$r < a$

$$\psi_{kra} = A \sinh(\kappa r) + B \cosh(\kappa r) = D e^{\kappa r}$$

c) $\left. \frac{\psi'}{\psi} \right|_{r=a} \rightarrow \frac{\kappa (\cos \delta j_0'(ka) - \sin \delta n_0'(ka))}{\cos \delta j_0(ka) - \sin \delta n_0(ka)} = \gamma_0 = \frac{\kappa e^{\kappa a}}{e^{ka}} = \kappa$

$$\tan \delta = \frac{\kappa j_0'(ka) - \gamma_0 j_0(ka)}{\kappa n_0'(ka) - \gamma_0 n_0(ka)} \quad \frac{a}{a} \Rightarrow \frac{x j_0'(x) - a \gamma_0 j_0(x)}{\kappa n_0'(x) - a \gamma_0 n_0(x)} =$$

$$= x \left(\frac{\cos x}{x} - \frac{\sin x}{x^2} \right) - a \gamma_0 \left(\frac{\sin x}{x} \right)$$

$$= \frac{\cos x - \frac{\sin x}{x} - a \gamma_0 \frac{\sin x}{x}}{\sin x + \frac{\cos x}{x} + a \gamma_0 \frac{\cos x}{x}}$$

$$\approx \frac{1 - \frac{x^2}{2} - 1 + \frac{x^2}{3} - a \gamma_0 (1 + \frac{x^2}{3})}{1 + \frac{x^3}{3} + \frac{1}{x} - \frac{1}{2} + a \gamma_0 (\frac{1}{x} - \frac{1}{2})} = \frac{-\frac{x^2}{2} + \frac{x^2}{3} - a \gamma_0 - a \gamma_0 \frac{x^2}{3}}{1 + \frac{1}{2} + \frac{x^3}{3} + \frac{1}{x} - \frac{1}{2} - a \gamma_0 \frac{1}{x}} = \frac{-a \gamma_0}{(a \gamma_0 - \delta)}$$

no long 1

1) cont.

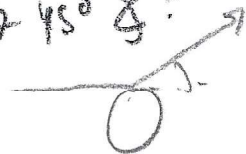
$$(1) \tan \delta = \frac{\cos k a - \frac{\sin k a}{k a} (1 + a \delta_0)}{\sin k a + \frac{\cos k a}{k a} (1 + a \delta_0)} \quad \delta_0 = \tan^{-1} \left(\frac{\cos k a - \frac{\sin k a}{k a} (1 + a \delta_0)}{\sin k a + \frac{\cos k a}{k a} (1 + a \delta_0)} \right)$$

d $V_0 \rightarrow \infty \quad \delta = \frac{2m(E - V_0)}{\hbar^2} \rightarrow \tan^{-1}(\infty) = \frac{\pi}{2} \checkmark$

it's shifted by $\pi/2$?

So like

hard scattering



e) $\sigma_0 = \frac{4\pi}{k^2} \sin^2(\delta_0) \quad V_0 \rightarrow 0$

$$\sigma_0 = \frac{4\pi}{k^2} \left(\delta_0 - \frac{\delta_0^3}{3!} \right)^2 = \frac{4\pi}{k^2} \delta_0^2 \quad \delta_0 \rightarrow 0$$

no potential, should be no phase shift

$$= 0$$

Nothing to hit, no cross section?

$$2) f_b = \frac{-2m}{q^2 \hbar^2} \int_0^\infty r V(r) \sin(qr) dr$$

$$= \left[\frac{-2m}{q^2 \hbar^2} \int_0^a r V_0 \sin(qr) dr \right] \cdot \frac{q}{q} \Rightarrow x = rq$$

$$= \frac{-2m}{q^2 \hbar^2} \int_0^{qa} V_0 x \sin(x) dx$$

$$\int_0^{qa} x \sin x = x \cos x \Big|_0^{qa} - \int_0^{qa} \cos(x) dx$$

$$= -qa \cos(qa) + \sin(qa)$$

$$f_b = \frac{-2m}{q^2 \hbar^2} V_0 (\sin(qa) - qa \cos(qa)) \quad \text{Amplitude}$$

$$d\sigma = |f_b|^2 d\Omega$$

$$\frac{d\sigma}{d\Omega} = |f_b|^2 = \frac{4m^2}{q^4 \hbar^4} V_0^2 (\sin(qa) - qa \cos(qa))^2$$